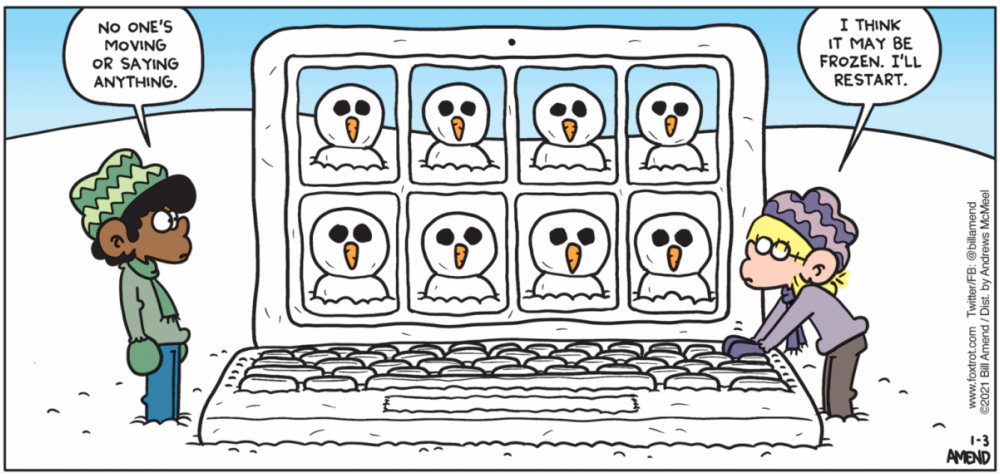
Autograded Homework #4: Memory and Caches, the CPU, Instruction Execution Cycle, and Performance

CS 255: Computer Architecture and Machine Level Programming



- To submit, complete the assignment on Gradescope.

1. Answer the following questions about memory and caches.

   (a) When the CPU reads from memory, which bus carries the actual bits of the data?

   A. Address bus
   B. Data bus True
   C. Control bus
   D. Instruction bus

   (b) What determines the maximum memory size the CPU can access?

   A. Width of the data bus
   B. Width of the address bus True
   C. CPU clock frequency
   D. Size of the ALU

   (c) What is the effect of a wider data bus?

   A. Fewer bytes per transfer, cheaper hardware
   B. More bytes per transfer, more expensive hardware True
   C. No effect on performance
   D. Reduces the number of memory addresses

   (d) What is the total size (in bytes) of `struct ABC` when stored in a 4-byte aligned memory?

```c
struct ABC {
    short a;
    int b;
    char c;
};
```

   12 = 2 bytes (short a) + 2 bytes padding + 4 bytes (int b) + 1 byte (char c) + 3 bytes padding

   (e) Which of the following memory access patterns demonstrates spatial locality?

   A. Repeatedly accessing the same array element.
   B. Accessing consecutive elements of an array. True
   C. Repeatedly accessing the same variable inside a loop.
   D. Alternating between two frequently used variables.

(f) Select all of the following statements about cache associativity that are true.

A. In a direct-mapped cache, each memory address maps to exactly one cache line. True

B. In a 2-way set-associative cache, a memory block can be stored in either of two lines within its assigned set. True

C. Increasing associativity always increases the number of cache lines.

D. Increasing associativity can reduce conflict misses. True

(g) Select all of the following statements about cache misses that are true.

A. A conflict miss occurs when multiple memory blocks compete for the same cache location. True

B. A conflict miss can occur even when the cache has unused space. True

C. A compulsory miss occurs the first time a memory block is accessed. True

D. A compulsory miss can be eliminated by increasing cache associativity.

2. Answer the following questions about CPU structure.

   (a) Which component performs arithmetic and logical operations?

   A. Register File
   B. ALU True
   C. Control Unit
   D. Program Counter

   (b) What is the primary responsibility of the Control Unit?

   A. Performing mathematical operations.
   B. Coordinating data flow and control signals among CPU components. True
   C. Storing operands and results.
   D. Managing input and output devices.

   (c) What does the Program Counter (PC) store?

   A. The current value of the ALU output.
   B. The address of the instruction currently being executed. True
   C. A copy of the last memory value read.
   D. The base address of the stack.

   (d) Where must data be stored for the ALU to perform an operation on it?

   A. Main memory
   B. Cache
   C. Register file True
   D. Control unit

3. Answer the following questions about logical operators.

(a) What does the bitwise AND (&) operator do?

A. Sets a bit to 1 if both inputs have 1 in that position. True
B. Sets a bit to 1 if either input has 1 in that position.
C. Flips each bit of its input.
D. Shifts bits to the right.

(b) If `a = 0b1100` and `b = 0b1010`, what is the result of `a^b` in binary? 0b0110

(c) Which operator would you use to flip the bits in a value?

A. &
B. |
C. ^
D. ~ True

```
int x = 0b00001001;
int y = set_bit(x, 4);
int z = clear_bit(y, 3);
```

(d) Referencing the code block, what is the binary value of `y` after calling `set_bit(x, 4)`? 0b00011001

(e) Referencing the code block, what is the decimal value of `z` after calling `clear_bit(y, 3)`? 17

4. Answer the following questions about the instruction execution cycle.

   (a) Place the following stages in the correct order (list A-E separated by spaces). C D A E B

   A. Execute
   B. Writeback
   C. Fetch
   D. Decode
   E. Memory Access

   (b) During which stage does the control unit generate the necessary signals to perform the instruction?

   A. Execute
   B. Writeback
   C. Fetch
   D. Decode True
   E. Memory Access

   (c) Which stage(s) access the register file? (list all that apply in alphabetical order separated by spaces) B E

   A. Fetch
   B. Decode
   C. Execute
   D. Memory Access
   E. Writeback

   (d) Why do some instructions (like add) effectively "skip" the memory stage?

   A. The instruction does not require reading or writing data from memory. True
   B. The CPU cannot access memory during execution.
   C. Memory access is handled during decode instead.
   D. The ALU replaces memory for simple operations.

5. Answer the following questions about CPU performance.

(a) A CPU runs at 3 GHz and executes one instruction per clock cycle. How many instructions can it execute per second? 3000000000 instructions

(b) A CPU runs at 2 GHz and executes one instruction per clock cycle. What is the total execution time (in ns) of a 1000 instruction program? 500 ns

(c) Which of the following statements about CPU A and CPU B are true if they run the same program?

- CPU A: 2 GHz, 1 CPI
- CPU B: 3 GHz, 2 CPI

A. CPU A executes fewer cycles per instruction than CPU B. True
B. CPU A completes the program in less time than CPU B. True
C. CPU A executes more cycles per second than CPU B.
D. Both CPUs complete the program in the same amount of time.

(d) Which change would most directly reduce program execution time?

A. Increasing clock frequency.
B. Reducing cycles per instruction.
C. Reducing number of instructions.
D. All of the above. True

(e) A program's execution time is dominated by memory access. Which improvement is most likely to help performance?

A. Increasing ALU speed.
B. Increasing clock frequency slightly.
C. Reducing memory access time. True
D. Increasing register file size.